

Corporate Finance — Illustrated Exam: Full Model Solutions

With Formula Rationale & Common Exam Pitfalls

Note: For every question this guide explains (1) which formula was used and **why**, (2) the full step-by-step working, (3) TI-84 Plus keystrokes, and (4) the most common mistakes students make on that type of question.

QUESTION TYPE 1 — Multiple Choice

MC Q1 — Inflation-Linked Annuity-Due + Growing Perpetuity

Answer: d. €2,550.93

Why this formula?

The product has **two phases**:

- **Phase 1:** Fixed payment of €50 at the beginning of each quarter for 3 years (12 quarterly payments starting immediately). This is an **annuity-due** — payments arrive at the start of each period.
- **Phase 2:** After 3 years (from quarter 13 onwards), payments grow at 0.5% per quarter forever. This is a **growing perpetuity**.

Because payments are **quarterly** but the rate is **annual**, you must convert the annual rate to a quarterly rate using the effective quarterly rate formula. The rate given is a **real** annual rate of 10%, which applies directly because the product is inflation-linked (payments grow with inflation).

STEP 1 – Convert annual real rate to quarterly effective rate

$$r_q = (1 + r_{\text{annual}})^{(1/4)} - 1 = (1.10)^{(0.25)} - 1 = 0.024114 = 2.4114\% \text{ per quarter}$$

STEP 2 – PV of Phase 1: Annuity-due, 12 payments of €50

$$\begin{aligned} \text{PV}_{\text{annuity}} &= C \times [(1 - (1+r_q)^{-n}) / r_q] \times (1 + r_q) \\ &= 50 \times [(1 - 1.024114^{-12}) / 0.024114] \times 1.024114 \\ &= 50 \times [0.24869 / 0.024114] \times 1.024114 \\ &= 50 \times 10.313 \times 1.024114 \\ &= €528.09 \end{aligned}$$

STEP 3 – PV of Phase 2: Growing perpetuity ($g = 0.5\%/quarter$)

First payment at $t = 12$ quarters: $C_{gp} = 50 \times 1.005 = €50.25$

Value at $t = 11$ (one period before first GP payment):

$$\text{PV}_{GP}(t=11) = C_{gp} / (r_q - g) = 50.25 / (0.024114 - 0.005) = 50.25 / 0.019114 = €2,629.01$$

Discount back to $t = 0$:

$$\text{PV}_{GP}(t=0) = 2,629.01 / (1.024114)^{11} = 2,629.01 / 1.30000 = €2,022.84$$

STEP 4 – Total price

$$P = 528.09 + 2,022.84 = €2,550.93 \quad \checkmark$$

TI-84 Plus Keystrokes

1. **Quarterly rate:** $1.10 \wedge 0.25 - 1$ ENTER → store: $\text{STO} \rightarrow X$ (0.024114)
2. **Annuity-due PV:** Finance → $N=12$, $I\%=2.4114$, $\text{PMT}=50$, $\text{FV}=0$, $\text{PV}=?$, then multiply by 1.024114 (annuity-due adjustment). Or: compute ordinary annuity $\text{PV} \times (1+r_q)$.
3. **GP at $t=11$:** $50.25 \div (X - 0.005)$ ENTER
4. **Discount to $t=0$:** $\div 1.024114^{11}$ ENTER
5. **Sum:** + Phase 1 result.

⚠ Common Pitfalls

Pitfall	What goes wrong
Using annual rate directly for quarterly payments	Massively understates the discount factor — always convert to matching period rate
Forgetting the annuity-due adjustment	Ordinary annuity formula gives PV at $t=-1$, not $t=0$; multiply by $(1+r)$
GP value placed at wrong time point	$PV = C/(r-g)$ gives value ONE period BEFORE first payment. First GP payment is at $t=12$, so GP value formula gives $t=11$, then discount 11 quarters back
Using nominal rate instead of real rate	For an inflation-linked product whose cash flows grow with inflation, use the real discount rate directly
Treating the growth phase as an ordinary annuity	It is perpetual (infinite), not for a fixed term

MC Q2 — Macaulay Duration

Answer: d. 1.95 years

Why this formula?

Macaulay duration measures the weighted-average time until you receive the bond's cash flows, where each weight is the share of the bond's present value received at that time. It is the go-to measure of a bond's interest-rate sensitivity (payback period).

FORMULA — Macaulay Duration

$$D_{\text{Mac}} = \frac{\sum [t \times PV(\text{Cft})]}{\text{Price}}$$

where $PV(\text{Cft}) = \text{Cft} / (1 + \text{YTM})^t$

Bond data: Maturity = 2 years, YTM = 3%, Coupon rate = 5% annually, Face = €100, Redeemed at par.

STEP 1 – Cash flows and their PVs

$$CF_1 = 5 \text{ (coupon)} \quad PV(CF_1) = 5 / 1.03 = €4.8544$$

$$CF_2 = 5 + 100 = 105 \text{ (coupon + par)} \quad PV(CF_2) = 105 / 1.03^2 = €98.9726$$

STEP 2 – Bond price

$$\text{Price} = 4.8544 + 98.9726 = €103.827$$

STEP 3 – Weighted time

$$\text{Numerator} = (1 \times 4.8544) + (2 \times 98.9726) = 4.8544 + 197.945 = €202.799$$

STEP 4 – Duration

$$D_{\text{Mac}} = 202.799 / 103.827 = 1.953 \text{ years} \approx 1.95 \text{ years} \quad \checkmark$$

TI-84 Plus Keystrokes

PV(CF1): $5 \div 1.03$ ENTER $\rightarrow 4.8544$
 PV(CF2): $105 \div 1.03 \times^2$ ENTER $\rightarrow 98.9726$
 Price : + (sum previous two) $\rightarrow 103.827$
 Num : $1 \times 4.8544 + 2 \times 98.9726$ $\rightarrow 202.799$
 Duration: $202.799 \div 103.827$ $\rightarrow 1.953$

⚠ Common Pitfalls

Pitfall	What goes wrong
Forgetting to include face value in the final cash flow	CF_n must be coupon + face value; omitting par underweights t=2 dramatically
Dividing numerator by face value instead of market price	Duration formula uses MARKET price in the denominator
Confusing Macaulay with Modified duration	Modified duration = Macaulay / (1+YTM); they are NOT the same
Using coupon rate instead of YTM to discount	Always discount with YTM (market rate), not the coupon rate

MC Q3 — DDM Stock Valuation (Gordon Growth Model)**Answer: b. €31.5**

Why this formula?

The company pays a stable dividend that grows at a constant rate → use the **Gordon-Shapiro (constant-growth DDM)**. Key insight: the growth rate g comes from the **plowback (retention) ratio × ROE**.

FORMULA – Gordon Growth Model

$$P_0 = D_1 / (r_e - g)$$

where:

D_1 = next expected dividend per share

r_e = cost of equity

g = sustainable growth rate = ROE × retention ratio = ROE × (1 – payout)

STEP 1 – EPS

$$\text{EPS} = \text{Net profit} / \text{Shares} = 3,600,000 / 2,000,000 = \text{€}1.80 \text{ per share}$$

STEP 2 – Next dividend (D_1)

$$D_1 = \text{payout} \times \text{EPS} = 0.35 \times 1.80 = \text{€}0.63$$

STEP 3 – Growth rate

$$g = \text{ROE} \times (1 - \text{payout}) = 12\% \times (1 - 0.35) = 12\% \times 0.65 = 7.8\% \approx 8\%$$

STEP 4 – Stock price

$$P_0 = D_1 / (r_e - g) = 0.63 / (10\% - 8\%) = 0.63 / 0.02 = \text{€}31.50 \quad \checkmark$$

Key convention used: Treat $D_1 = \text{payout} \times \text{EPS}_0$ (the current year's declared dividend) and round g to 8%, consistent with the exam's intended precision.

TI-84 Plus Keystrokes

$$\text{EPS: } 3600000 \div 2000000 = 1.80$$

$$\text{D1: } 0.35 \times 1.80 = 0.63$$

$$g: 0.12 \times 0.65 = 0.078 \rightarrow \text{round to } 0.08$$

$$\text{P0: } 0.63 \div (0.10 - 0.08) = 0.63 \div 0.02 = 31.50$$

⚠ Common Pitfalls

Pitfall	What goes wrong
Using ROE as the growth rate directly	$g = \text{ROE} \times \text{retention ratio}$, not $g = \text{ROE}$
Forgetting $(1 - \text{payout})$ to get retention ratio	$\text{payout} = 35\% \rightarrow \text{retention} = 65\%$, not 35%
Using $r - g$ with unrounded g (7.8%)	$0.63 / 0.022 \approx 28.6$; exam uses $g \approx 8\%$ to get 31.5
Confusing D_0 with D_1	The Gordon model uses D_1 (next year's dividend), but here the declared dividend is treated as D_1 directly
Using book value of shares instead of number of shares	Divide profit by NUMBER of shares, not by book value

MC Q4 — Portfolio Sharpe Ratio

Answer: b. 0.82

Why this formula?

The **Sharpe Ratio** measures risk-adjusted return — how much excess return (above the risk-free rate) you earn per unit of total portfolio risk (standard deviation). It requires computing the portfolio's expected return and standard deviation first.

FORMULAS — 2-Asset Portfolio

$$E(R_p) = w_A \times E(R_A) + w_B \times E(R_B)$$

$$\sigma_p^2 = w_A^2 \times \sigma_A^2 + w_B^2 \times \sigma_B^2 + 2 \times w_A \times w_B \times \rho_{AB} \times \sigma_A \times \sigma_B$$

$$\text{Sharpe Ratio} = [E(R_p) - r_f] / \sigma_p$$

Data: $w_A = 35\%$, $w_B = 65\%$, $E(R_A) = 12\%$, $\sigma_A = 15\%$, $E(R_B) = 15\%$, $\sigma_B = 20\%$, $\rho = 0.6$, $r_f = 0.25\%$

STEP 1 – Expected portfolio return

$$E(R_p) = 0.35 \times 12\% + 0.65 \times 15\% = 4.20\% + 9.75\% = 13.95\%$$

STEP 2 – Portfolio variance

$$\begin{aligned}\sigma^2_p &= (0.35)^2 \times (0.15)^2 + (0.65)^2 \times (0.20)^2 + 2 \times 0.35 \times 0.65 \times 0.6 \times 0.15 \times 0.20 \\ &= 0.1225 \times 0.0225 + 0.4225 \times 0.04 + 2 \times 0.35 \times 0.65 \times 0.6 \times 0.03 \\ &= 0.0027563 + 0.016900 + 0.008190 \\ &= 0.027846\end{aligned}$$

STEP 3 – Portfolio standard deviation

$$\sigma_p = \sqrt{0.027846} = 16.69\%$$

STEP 4 – Sharpe Ratio

$$\text{Sharpe} = (13.95\% - 0.25\%) / 16.69\% = 13.70\% / 16.69\% = 0.821 \approx 0.82 \quad \checkmark$$

⚠ Common Pitfalls

Pitfall	What goes wrong
Taking weighted average of individual SDs	$\sigma_p \neq w_A \times \sigma_A + w_B \times \sigma_B$; you MUST use the variance formula with the covariance term
Forgetting the cross-product term	The $2 \times w_A \times w_B \times \rho \times \sigma_A \times \sigma_B$ term is the diversification effect — leaving it out overstates risk
Subtracting book value instead of risk-free rate	$\text{Sharpe} = (R_p - \mathbf{rf}) / \sigma_p$, not $(R_p - 0) / \sigma_p$
Computing σ^2 in % but then using σ = variance directly	Take the square root of variance to get standard deviation
Wrong weights (reversing A and B)	Re-read carefully which asset has which weight

MC Q5 — Portfolio Risk (Variance)

Answer: c. variance = 3.936%

Why this formula?

Same 2-asset portfolio variance formula. The question asks "which estimate **best quantifies** portfolio risk?" — this is a conceptual trap. In finance, **variance** is the primary measure of risk (not standard deviation, which is derived from it), and the answer choices deliberately mix variance and standard deviation to test whether you can compute both correctly.

Data: $w_A = 35\%$, $w_B = 65\%$, $\sigma_A = 20\%$, $\sigma_B = 22\%$, $\rho = 0.7$

STEP 1 – Portfolio variance

$$\sigma_p^2 = (0.35)^2 \times (0.20)^2 + (0.65)^2 \times (0.22)^2 + 2 \times 0.35 \times 0.65 \times 0.7 \times 0.20 \times 0.22$$

$$= 0.1225 \times 0.04 + 0.4225 \times 0.0484 + 2 \times 0.2275 \times 0.7 \times 0.044$$

$$= 0.004900 + 0.020449 + 0.014014$$

$$= 0.039363$$

Expressed as a percentage: 3.936% ✓

STEP 2 – Standard deviation (to check wrong options)

$$\sigma_p = \sqrt{0.039363} = 19.84\%$$

→ Neither 15.19% nor 25.19% is correct.

→ Variance = 3.936% ✓, Variance = 4.936% ✗

Note on "%" notation for variance: When variance is expressed as a percentage, it means the decimal value $\times 100$. So variance = 0.03936 is stated as "3.936%." This is a notational convention that often confuses students.

⚠ Common Pitfalls

Pitfall	What goes wrong
Reporting σ instead of σ^2	If question asks for variance, don't give standard deviation
Computing σ and calling it "variance = 19.84%"	19.84% is the standard deviation , not variance
Misreading the correlation as covariance	$\rho = 0.7$ is the correlation; covariance = $\rho \times \sigma_A \times \sigma_B$
Picking the "nearest" answer without calculating	Always compute; you need to distinguish 3.936% from 4.936%

QUESTION TYPE 2 — Short Closed Questions

Q2.1 — Itsme Acquiring NextAuth: Financial Valuation

Model Answer:

The acquisition of NextAuth should be analysed as a **Net Present Value (NPV) problem** from the acquirer's perspective. The key financial valuation elements are:

1. **Standalone value of NextAuth:** Estimated using a **DCF model** — project NextAuth's future free cash flows and discount them at an appropriate risk-adjusted discount rate (WACC or cost of equity). This gives the intrinsic value of the target as an independent entity.
2. **Synergy value:** Acquisitions often create synergies (cost savings, revenue gains, tax benefits). These expected synergies must be quantified and added to the standalone value.
3. **Acquisition price (premium):** The acquirer typically pays a premium above the current market value (or DCF value). The NPV of the acquisition is:

$$\text{NPV}_{\text{acquisition}} = (\text{Standalone Value} + \text{Synergies}) - \text{Acquisition Price}$$

1. **The decision rule:** Proceed with the acquisition **if and only if NPV > 0** — meaning the combined value exceeds the price paid. Overpaying destroys shareholder value even if the target itself is a good company.

Q2.2 — PVGO: Present Value of Growth Opportunities

Model Answer:

What PVGO is: PVGO is the portion of a company's stock price that reflects the value of its **future investment opportunities** (growth) rather than its current earnings capacity. It represents how much investors are paying for expected profitable future projects.

$$\text{PVGO} = P_0 - \text{EPS} / r_e$$

(Stock price) – (Value as zero-growth perpetuity)

Why PVGOs arise: PVGOs arise because a company can reinvest earnings at a **return on equity (ROE) that exceeds the cost of equity (r_e)**. If $\text{ROE} > r_e$, retained earnings generate more value than if they were paid out, creating positive PVGO. If $\text{ROE} = r_e$, retention adds no value ($\text{PVGO} = 0$). If $\text{ROE} < r_e$, the company should pay out all earnings.

How to estimate it: 1. Compute P_0 using the Gordon Growth Model (which includes growth) 2. Compute the no-growth benchmark: EPS / r_e 3. $\text{PVGO} = P_0 - \text{EPS} / r_e$

Q2.3 — Three Drivers of Bond Volatility

Model Answer:

Bond **volatility** (= modified duration) measures how much a bond's price changes when interest rates change by 1%. Three factors drive it:

1. Time to maturity: Longer maturity → higher volatility. A longer-maturity bond has more cash flows far in the future, which are highly sensitive to discount rate changes. A bond maturing in 20 years has far more interest rate risk than one maturing in 1 year.

2. Coupon rate: Lower coupon rate → higher volatility. A lower coupon bond delivers more of its value through the final (distant) principal repayment, making it more back-loaded and thus more sensitive to rate changes. A zero-coupon bond has the highest volatility for its maturity.

3. Yield-to-maturity (YTM) level: Lower YTM → higher volatility. The relationship between price and yield is convex (non-linear). At low yield levels, the same change in yield causes a larger absolute price change than at high yield levels.

$$\text{Modified Duration (Volatility)} = \text{Macaulay Duration} / (1 + \text{YTM})$$

Q2.4 — CML vs SML: What Do They Tell Us?

Model Answer:

$$\text{CML: } E(R_p) = r_f + [(E(R_m) - r_f) / \sigma_m] \times \sigma_p$$

$$\text{SML: } E(R_i) = r_f + \beta_i \times [E(R_m) - r_f]$$

CML (Capital Market Line): The CML describes the **risk-return tradeoff for efficiently diversified portfolios** — portfolios that combine the risk-free asset with the market portfolio. It tells you: for each unit of total risk (σ_p) you take on, you are compensated by a proportional expected return above r_f . The slope $(E(R_m) - r_f) / \sigma_m$ is the "price of total risk." Applies only to: portfolios on the efficient frontier.

SML (Securities Market Line): The SML describes the **required return for ANY individual asset or portfolio** as a function of its systematic risk (beta), not total risk. The key insight is that in a diversified market, only **systematic risk** (risk that cannot be diversified away) commands a return premium. Unsystematic risk is irrelevant to pricing because rational investors diversify it away for free.

Core message: The two lines share the same intercept (rf) and the same market portfolio point. The CML uses total risk; the SML uses only systematic risk. The SML is the correct pricing model for individual securities because investors are only rewarded for bearing **undiversifiable** risk.

Q2.5 — Diversification

Model Answer:

How diversification arises: When you combine assets in a portfolio, their returns do not move perfectly together (correlation < 1). When one asset falls, another may hold steady or rise. As a result, the portfolio's volatility is lower than the weighted average of individual volatilities. The lower the correlation between assets, the greater the reduction in risk.

$$\sigma_p < w_A \times \sigma_A + w_B \times \sigma_B \quad \text{when } \rho_{AB} < 1$$

What type of risk is eliminated: Diversification eliminates **unsystematic (idiosyncratic / company-specific) risk** — the portion of a stock's volatility caused by factors unique to that company (management decisions, product recalls, lawsuits, etc.).

What cannot be eliminated: Systematic (market) risk — risk caused by economy-wide factors (interest rate changes, recessions, pandemics) — affects all assets simultaneously and cannot be reduced by diversification. This is why beta (systematic risk) is what the market prices; investors get no reward for taking unsystematic risk because it is freely diversifiable.

QUESTION TYPE 3 — Large Exercise: SRB NV

Q3a — Cost of Capital (WACC)

Why WACC?

WACC is the appropriate discount rate for an investment project when the project has the same risk profile as the company overall. It represents the **minimum required return** that a project must generate to create value for all capital providers (both equity holders and debt holders, after tax). Using WACC ensures you account for the tax deductibility of interest payments.

FORMULA – WACC

$$WACC = (E/V) \times r_e + (D/V) \times r_d \times (1 - T)$$

where:

- E = market value of equity
- D = market value of debt
- V = E + D (total capital, market value)
- r_e = cost of equity (from CAPM)
- r_d = cost of debt
- T = corporate tax rate

STEP 1 – Compute Beta via CAPM inputs

$$\begin{aligned}\beta &= \text{Cov}(R_{\text{SRB}}, R_{\text{market}}) / \text{Var}(R_{\text{market}}) \\ &= 0.04 / (0.14)^2 \\ &= 0.04 / 0.0196 \\ &= 2.041\end{aligned}$$

Interpretation: SRB is about twice as volatile as the market. A 1% market rise leads to a ~2% rise in SRB's share price.

STEP 2 – Cost of Equity via CAPM

$$\begin{aligned}r_e &= r_f + \beta \times (E(R_m) - r_f) \\ &= 5\% + 2.041 \times (9\% - 5\%) \\ &= 5\% + 2.041 \times 4\% \\ &= 5\% + 8.163\% \\ &= 13.16\%\end{aligned}$$

Interpretation: SRB equity holders require a 13.16% annual return to compensate for the risk they bear.

STEP 3 – Capital structure weights (always use MARKET values)

$$\begin{aligned}\text{Market value of equity } E &= 5,000,000 \text{ shares} \times \text{€}8 = \text{€}40,000,000 \\ \text{Market value of debt } D &= \text{€}60,000,000 \\ \text{Total } V &= 40\text{M} + 60\text{M} = \text{€}100,000,000\end{aligned}$$

$$\begin{aligned}w_E &= 40\text{M} / 100\text{M} = 0.40 \quad (40\% \text{ equity}) \\ w_D &= 60\text{M} / 100\text{M} = 0.60 \quad (60\% \text{ debt})\end{aligned}$$

STEP 4 – WACC

$$\begin{aligned}WACC &= 0.40 \times 13.16\% + 0.60 \times 7\% \times (1 - 0.30) \\ &= 5.264\% + 0.60 \times 4.90\% \\ &= 5.264\% + 2.940\% \\ &= 8.20\%\end{aligned}$$

Interpretation: SRB must earn at least 8.20% per year on any new investment to create value. This is the hurdle rate for the project evaluation below.

Role of WACC in investment analysis: The WACC is used as the discount rate in NPV calculations. If a project's internal rate of return (IRR) > WACC, it creates

value. If the project's NPV discounted at WACC is positive, accepting it increases firm value.

⚠ Common Pitfalls

Pitfall	What goes wrong
Using book value weights instead of market value	Book value is historical; only market value reflects true economic weight
Forgetting $(1 - T)$ on the cost of debt	Interest is tax-deductible; after-tax cost = $r_d \times (1 - T)$
Using the risk-free rate or coupon rate as r_d	Use the stated cost of debt (7%), not the government bond rate
Confusing Cov with Var in the beta formula	$\beta = \text{Cov} / \text{Var}(\text{market})$, NOT $\text{Cov} / \text{SD}(\text{market})$

Q3b — Project Choice: EAA Method

Why EAA (Equivalent Annual Annuity)?

SRB is a **going concern** — after Proposal A's machinery wears out after 4 years, SRB must reinvest again. Similarly for Proposal B after 6 years. Because the two proposals have **different economic lives** (4 vs. 6 years) and will be repeated, a simple NPV comparison is misleading. NPV is affected by the length of the investment horizon.

The **EAA method** converts each NPV into a per-year annualized equivalent, allowing apples-to-apples comparison regardless of lifespan.

FORMULA — Equivalent Annual Annuity

$$\text{EAA} = \text{NPV} / \text{AF}(r, n)$$

where $\text{AF}(r, n) = [1 - (1+r)^{-n}] / r$ (annuity factor)

Decision rule: Choose the proposal with the higher EAA.

Assumption: Both machines are replaced by identical projects at end of life (going-concern replacement chain assumption).

Operating Cash Flow Formula:

$$OCF = (\Delta \text{Revenue} - \Delta \text{Costs}) \times (1 - T) + \text{Depreciation} \times T$$

This avoids having to compute EBIT and NOPAT separately.

Alternatively:

$$OCF = \text{NOPAT} + \text{Depreciation} = [\text{EBIT} \times (1-T)] + \text{Dep}$$

Proposal A (4 years, CAPEX = €400,000)

$$\text{Annual depreciation} = 400,000 / 4 = \text{€}100,000 \text{ (straight-line)}$$

$$\Delta \text{Revenue} = \text{€}175,000$$

$$\Delta \text{Operating costs} = \text{€}40,000$$

$$\Delta \text{EBIT} = 175,000 - 40,000 - 100,000 = \text{€}35,000$$

$$\text{Tax (30\%)} = 35,000 \times 0.30 = \text{€}10,500$$

$$\text{NOPAT} = 35,000 - 10,500 = \text{€}24,500$$

$$OCF = \text{NOPAT} + \text{Dep} = 24,500 + 100,000 = \text{€}124,500 \text{ per year}$$

Annuity Factor at WACC = 8.20%, n = 4:

$$AF(8.20\%, 4) = [1 - (1.082)^{-4}] / 0.082$$

$$= [1 - 0.7296] / 0.082 = 0.2704 / 0.082 = 3.297$$

$$NPV_A = -400,000 + 124,500 \times 3.297$$

$$= -400,000 + 410,481$$

$$= \text{€}10,481$$

$$EAA_A = NPV_A / AF(8.20\%, 4) = 10,481 / 3.297 = \text{€}3,179 \text{ per year}$$

Proposal B (6 years, CAPEX = €500,000)

$$\text{Annual depreciation} = 500,000 / 6 = \text{€}83,333 \text{ (straight-line)}$$

$$\Delta \text{Revenue} = \text{€}190,000$$

$$\Delta \text{Operating costs} = \text{€}40,000 + \text{€}5,000 \text{ (maintenance)} = \text{€}45,000$$

$$\Delta \text{EBIT} = 190,000 - 45,000 - 83,333 = \text{€}61,667$$

$$\text{Tax (30\%)} = 61,667 \times 0.30 = \text{€}18,500$$

$$\text{NOPAT} = 61,667 - 18,500 = \text{€}43,167$$

$$OCF = 43,167 + 83,333 = \text{€}126,500 \text{ per year}$$

Annuity Factor at WACC = 8.20%, n = 6:

$$AF(8.20\%, 6) = [1 - (1.082)^{-6}] / 0.082$$

$$= [1 - 0.6231] / 0.082 = 0.3769 / 0.082 = 4.596$$

$$NPV_B = -500,000 + 126,500 \times 4.596$$

$$= -500,000 + 581,394$$

$$= \text{€}81,394$$

$$EAA_B = NPV_B / AF(8.20\%, 6) = 81,394 / 4.596 = \text{€}17,710 \text{ per year}$$

Decision

	Proposal A	Proposal B
CAPEX	€400,000	€500,000
Economic life	4 years	6 years
Annual OCF	€124,500	€126,500
NPV	€10,481	€81,394
EAA	€3,179/yr	€17,710/yr

→ **Choose Proposal B** ($EAA_B = €17,710 \gg EAA_A = €3,179$)

Both proposals create positive value (both EAAs > 0), but Proposal B creates approximately **5.6× more annualized value** per year. Despite the higher upfront cost, Proposal B's 6-year life and higher incremental cash flows generate far superior returns.

TI-84 Plus — Finance Menu for NPV

Press: 2ND → FINANCE (or APPS → Finance)
 Select: npv(
 Syntax: npv(rate, initial investment as negative, {annual CF list})
 → npv(8.20, -400000, {124500, 124500, 124500, 124500})
 For EAA, use:
 → pmt after solving $PV = NPV$, N = project life, $I\%$ = WACC, $FV = 0$

⚠ Common Pitfalls

Pitfall	What goes wrong
Comparing NPVs directly for different-life projects	Ignores that A must be replaced after 4 years; NPV_A covers only 4 years, NPV_B covers 6
Forgetting to deduct maintenance cost in Proposal B	Extra €5,000/year reduces EBIT — easy to overlook in the problem
Not adding back depreciation to get OCF	Depreciation is non-cash; it reduces tax but must be added back to get actual cash flow
Using book value of equity in WACC	Always use market value (shares × market price)
Choosing A because it has lower upfront cost	The correct decision metric is EAA, not CAPEX size

Q3c — Inventory Increase → Working Capital Effect

Answer: YES, this has an impact on financial value — it reduces NPV.

Explanation: An increase in inventory = an increase in **Net Working Capital (NWC)**. NWC changes represent cash tied up in operations. When inventory rises, the firm must pay for goods or raw materials before it can sell them — this is a real cash outflow.

- At the **start of the project**, additional NWC required = cash outflow → **reduces the initial investment base** → lowers NPV.
- At the **end of the project**, NWC is typically recovered (inventory sold off, receivables collected) → cash inflow.

Because the cash outflow occurs NOW ($t=0$) and the recovery occurs later (at project end), the **TIME VALUE OF MONEY** means the initial outflow has higher PV than the future recovery — **net effect is negative on NPV**.

Rule: Any increase in NWC reduces NPV. Any decrease in NWC (e.g., more efficient inventory in Q3c of the mock exam) increases NPV.

Q3d — Declining Balance Depreciation vs Straight-Line

Answer: YES, this has an impact on financial value — it increases NPV.

Explanation: Depreciation is a **tax shield** (it reduces taxable income and thus taxes paid). The total amount of tax shielded over the project life is the same under both methods (same total depreciation = CAPEX). However, timing differs:

- **Straight-line:** Equal tax shields spread evenly over the project life.
- **Declining balance (degressive):** Larger depreciation deductions in early years → larger tax shields early → smaller later.

Because of the **time value of money**, early cash inflows are worth more than later ones. Under declining balance, the firm receives the tax benefits sooner, so the **PV of the tax shield is higher** → **NPV is higher** under declining balance depreciation.

Conclusion: Declining balance depreciation increases the NPV of a project compared to straight-line, even though the total depreciation amount is identical.

QUESTION TYPE 4 — ACTUA & Capita Selecta

Q4.1 — GameStop (Feb 2021): Short Squeeze

Part a — Story with key terms

Short position: Institutional investors (hedge funds, notably Melvin Capital) had taken **short positions** in GameStop — they borrowed shares, sold them at the current market price (~€4), and planned to buy them back cheaper when the stock fell, pocketing the difference. This bet was based on GameStop's declining retail business model.

Short squeeze: In January 2021, retail investors on Reddit's WallStreetBets forum coordinated a massive buy campaign in GameStop. As the stock price surged (from ~\$20 to ~\$480), short-sellers faced enormous losses. To limit losses, they had to **buy back the shares** they had shorted, which drove the price even higher — a **short squeeze**: a self-reinforcing cycle where short-covering creates more upward price pressure, forcing more short-covering.

Margin requirements: Brokers require short-sellers to post collateral (margin) proportional to their position size and current losses. As GameStop's price soared, the value of the short positions ballooned to catastrophic losses, triggering **margin calls** — demands for additional collateral. Unable to meet these requirements, hedge funds were forced to buy shares immediately at any price, accelerating the squeeze.

Part b — EMH Implications

Efficient Market Hypothesis (EMH): EMH states that asset prices at all times reflect all available information. In its forms: - **Weak:** Prices reflect all historical trading data - **Semi-strong:** Prices reflect all publicly available information - **Strong:** Prices reflect all public AND private information

GameStop's challenge to EMH: GameStop's fundamental value (based on cash flows, earnings) did not justify a price of \$480. The price explosion was driven by coordinated retail investor action — a social phenomenon, not new information about the company's future cash flows. This challenges the **semi-strong** form of EMH: if markets were efficient, the crowd-driven buying should not have pushed prices so far above fundamental value.

It also provides real-world evidence of **behavioral finance** concepts: herding behavior, momentum trading, and the power of social media to coordinate market movements that deviate from rational pricing. The episode shows that markets

can be inefficient in the **short run**, even if they tend toward efficiency in the long run.

Q4.2 — Long Call Option

Part a — Long Call position and asymmetric payoff

What it is: A **long call** is the purchase of a call option. It gives the holder the **right (but NOT the obligation)** to BUY an underlying asset at a pre-agreed **strike price (K)** on or before the expiration date. To acquire this right, the buyer pays an upfront **premium (P)**.

How the asymmetric payoff arises:

At maturity, two scenarios:

If $S_T > K$: Exercise! Payoff = $S_T - K$. Profit = $(S_T - K) - P$

If $S_T \leq K$: Don't exercise. Loss = P (the premium paid)

Maximum loss: $-P$ (limited, known upfront)

Maximum gain: Unlimited (as $S_T \rightarrow \infty$, profit $\rightarrow \infty$)

This creates the characteristic "hockey stick" profile: - To the LEFT of the strike: flat line at $-P$ (you let the option expire, lose only what you paid) - To the RIGHT of the strike: profit rises one-for-one with the stock price (payoff line), but profit (net of premium) starts positive only once $S_T > K + P$ (the break-even point)

The payoff line (gross) always sits Premium above the profit line (net).

Part b — What risk does a Long Call hedge?

A long call hedges **upside price risk** — the risk that the price of an asset you **need to buy** will rise in the future.

Mechanism: If you need to acquire an asset in 6 months and fear the price will rise sharply, buying a call option today locks in the maximum purchase price (the strike). If prices do rise, your option compensates. If prices fall, you simply don't exercise the option and buy at the (lower) market price.

Specific corporate example:

An airline company needs to purchase large quantities of jet fuel continuously. It fears a rise in oil prices, which would increase operating costs and squeeze margins. The airline buys **call options on crude oil (or jet fuel)**. If fuel prices spike above the strike price, the airline exercises the

options and effectively pays the strike price. If prices stay low, it lets the options expire (losing only the premium) and buys fuel at the favourable market price. Result: The airline's maximum fuel cost is capped, providing budget certainty.

Other valid examples: - An importer buying USD in 6 months (buys a USD call to cap exchange rate) - A manufacturer buying copper futures (buys call to cap raw material cost)

SUMMARY CHEAT SHEET — Formulas & When to Use Them

Situation	Formula	Key condition
Fixed payment, beginning of each period	Annuity-due PV = ordinary annuity $\times (1+r)$	Payments at START of period
Fixed payment, end of each period	Ordinary annuity PV = $C \times [1 - (1+r)^{-n}] / r$	Payments at END of period
Payments grow forever at rate g	Growing perpetuity PV = $C / (r-g)$	$r > g$ always required
Bond price	$P = \sum [CF_t / (1+YTM)^t]$	Discount each CF at YTM
Bond duration	$D = \sum [t \times PV(CF_t)] / \text{Price}$	Weighted-average time
DDM constant growth	$P = D_1 / (r-g)$, $g = ROE \times (1 - \text{payout})$	Steady-state growth
PVGO	$PVGO = P_0 - EPS / r_e$	Only if $ROE > r_e$
Beta	$\beta = \text{Cov}(R_i, R_m) / \text{Var}(R_m)$	Use market-value data
CAPM (COE)	$r_e = r_f + \beta(R_m - r_f)$	Systematic risk only
WACC	$w_E \times r_e + w_D \times r_d \times (1-T)$	Market-value weights
NPV	$-CAPEX + \sum [OCF_t / (1+r)^t]$	$r = \text{WACC}$
OCF	$(\text{Rev} - \text{Costs}) \times (1-T) + \text{Dep} \times T$	Add back dep tax shield
EAA	$EAA = NPV / AF(r,n)$	For different-life projects
Sharpe Ratio	$(E(R_p) - r_f) / \sigma_p$	Total risk (σ), not β
Portfolio variance	$w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho \sigma_A \sigma_B$	2-asset formula

Solutions prepared using course material from CH3-CH10 and ACTUA cases. All numerical results verified with Python.